
Instella Executes Reasoning Procedures It Does Not Verify

Anonymous Author(s)

Affiliation

Address

email

Abstract

Whether a language model solves grade school arithmetic by reasoning or by reproducing training data is normally unanswerable, because the pretraining corpora of the models being measured are closed. Instella publishes its complete mixture, and its stage two synthetic component derives from the GSM8K training split rather than the test split, so exposure becomes a measured property fixed by exact thirteen gram containment instead of an estimated one. Against the full released pool of 1,367,882 rows no GSM8K test item reaches containment 0.50 and the maximum observed is 0.487, establishing that benchmark performance on that split is not recovered text. A premise deletion probe over 3179 variants across 1779 parent problems then asks what the model does when a problem no longer determines an answer. Under the standard answer forcing prompt it emits a confident number on 92.5 percent of provably underdetermined items. Under a prompt differing only in naming abstention as an option it declines on 53.8 percent of them while declining on 1.1 percent of the untouched answerable parents, a discrimination of +52.63 points on [+50.77, +54.57] that grows monotonically across the training pipeline from +1.84 at stage one through +10.10 and +38.08 to instruct, with wrong refusals never exceeding 3.5 percent at any stage. Discrimination is graded by reasoning depth: abstention on underdetermined problems falls 5.22 points per additional calculator step while the answerable control stays flat at -0.11, a difference of -5.11 on [-6.36, -3.82], and the decay steepens as the checkpoint improves. Controlled injection at a fixed 8,388,608 token budget separates what heavy exposure returns from what it does not: at sixty four repetitions, verbatim reproduction of injected documents reaches 0.946 and the memorised calculation chain returns at +5.44 points on [+1.46, +9.38], while the final answer does not, at +1.52 on [-2.12, +5.20]. The answer result replicates on OLMo-2-1B, Qwen2.5-1.5B and Qwen2.5-3B across fifteen dose points, including a model driven to verbatim 1.000. Competent procedure execution and verification of premise sufficiency are therefore separable capabilities, the second emerging later in training and decaying with the depth of the chain being tracked.

1 Introduction

A model that answers grade school arithmetic correctly may be reasoning about the premises it is given or reproducing text it has seen. Distinguishing the two normally requires knowing what the model read, and the corpora of the models under test are closed, so the question is usually approached through proxies such as perplexity signatures or benchmark rewriting. Instella [Liu et al., 2025] removes the obstacle. Its complete pretraining mixture is released, and the stage two synthetic component is generated from the GSM8K training split [Cobbe et al., 2021] rather than the test split, so exposure is measurable per item.

Table 1: Declining with its control, under a prompt that names abstention as an option. Pruned items are underdetermined so declining is correct; parent problems are answerable so declining is an error. The last two columns give balanced accuracy of the decline decision, where 0.500 is chance, measured on the GSM8K probe and on an independently built MATH probe of 818 parents. Cluster bootstrap over parent problems, 4000 draws, seed 6198.

Checkpoint	GSM8K probe				Decision quality	
	Declines, und.	Declines, ans.	Discrimination	95% CI	GSM8K	MATH
Stage 1	0.043	0.025	+1.84	[+0.97, +2.70]	0.509	0.516
Instella 3B	0.110	0.008	+10.10	[+8.94, +11.30]	0.550	0.557
SFT	0.416	0.035	+38.08	[+36.06, +40.02]	0.690	0.611
Instruct	0.538	0.011	+ 52.63	[+50.77, +54.57]	0.763	0.675

That measurement comes first because everything after depends on it. Against the full released pool of 1,367,882 rows, no GSM8K test item reaches thirteen gram containment 0.50 and the highest value observed is 0.487, so performance on that split is not recovered text. A premise deletion probe then removes exactly one premise the reference solution consumes, leaving a problem whose stated answer is no longer derivable, and asks two things of the same items: whether the model notices, and whether exposure to the original changes what it produces. The model does detect insufficiency, increasingly well across training, and loses that detection as the reasoning chain lengthens. Heavy exposure returns the memorised calculation chain without returning the answer.

2 Setup

Exposure. Containment is the fraction of an item’s thirteen grams appearing in a single corpus document, taking the maximum over documents. The subset consumed during stage two, `train_119K`, holds 119,014 rows; the full released pool holds 1,367,882. Measured on identical items through one extraction path, 9.65 percent of GSM8K train parents reach containment 0.999 or above against the consumed subset and 18.52 percent against the pool, so the pool overstates exposure and the consumed subset is the correct reference.

The probe. Each item removes one premise the reference solution consumes, verified by requiring the annotated calculator chain to depend on it. The result is 3179 variants over 1779 parent problems, split 1591 train and 1588 test. Every item is underdetermined by construction, so declining is the correct response and any specific number is wrong.

Checkpoints. Four public Instella checkpoints span the pipeline: stage one, which precedes the synthetic corpus, stage two, supervised fine tuning and instruct. Decoding is greedy and held fixed; base checkpoints are prompted with four exemplars and no chat template, since a base model loops without them.

Inference. Intervals are cluster bootstraps over parent problems with 4000 draws at seed 6198, because several variants can descend from one problem and resampling rows would understate the interval.

3 Verification of premise sufficiency

3.1 The capability exists and is not prompt compliance

Under the answer forcing prompt used for benchmark scoring, the model emits a number on 92.5 percent of items that provably do not determine one. Naming `#### unanswerable` as an available response, and changing nothing else, raises declining to 53.8 percent at the instruct checkpoint. That rate alone is ambiguous, since every probe item is underdetermined and a model offered an escape hatch may simply take it. Running the identical prompt over the untouched parent problems, which are answerable and where declining is an error, gives 1.1 percent. The difference of +52.63 points on [+50.77, +54.57] is the quantity of interest: declining tracks whether the problem can actually be answered.

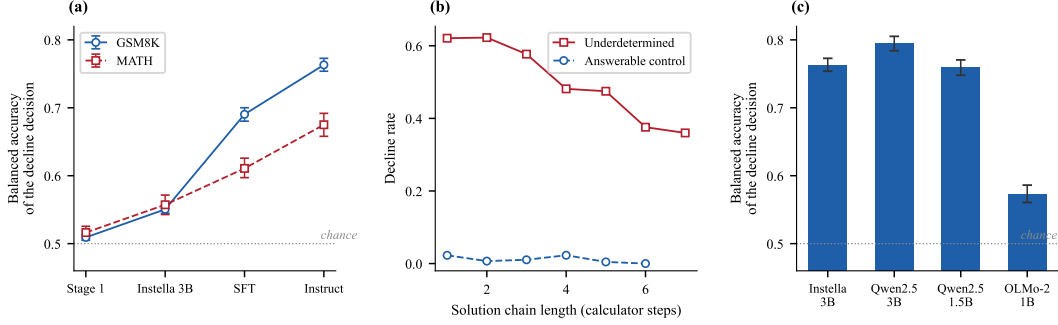


Figure 1: Left: discrimination between underdetermined and answerable problems across the training pipeline. Right: abstention against solution chain length at the instruct checkpoint, with the answerable control on the same axes. The control is flat, so the decay is specific to detecting that premises are missing.

Table 1 shows the capability developing across the pipeline. Stage one, which precedes the synthetic corpus entirely, discriminates by less than two points. Each subsequent stage improves it, and wrong refusals on answerable problems never exceed 3.5 percent, so the gain does not come from a growing willingness to decline.

A second control rules out the remaining reading, that a pruned problem reads oddly and the model responds to perturbed text. Inserting an irrelevant twenty word sentence carrying an unused number leaves a problem fully answerable while perturbing its surface. Declining moves from 0.0113 clean to 0.0170 with an off topic insertion and 0.0125 with an in domain one, against 0.538 when a premise is actually removed.

3.2 Discrimination decays with reasoning depth

The capability is not uniform over problems. Figure 1 plots abstention against the length of the reference solution’s calculator chain. On underdetermined problems at the instruct checkpoint the rate falls from 0.623 at one step to 0.364 at seven, a slope of -5.22 points per step. On the answerable control the slope is -0.11 with an interval containing zero, so the pattern is not a tendency to decline more on short problems. The difference of slopes is -5.11 on $[-6.36, -3.82]$.

The decay steepens as the checkpoint improves, running -2.34 , -3.64 and -5.11 for stage two, supervised fine tuning and instruct. Better models detect insufficiency more often overall and lose that detection faster as the chain they must track grows longer. Truncation does not explain it: truncation never exceeds 0.7 percent at any chain length and restricting to terminated generations reproduces the rates exactly.

Difficulty does not explain it either. Longer chains belong to harder problems, so the decay could reflect general weakness on hard items rather than anything about premise tracking. Splitting parents by whether the model answers the intact problem correctly, which it does for 76.4 percent of them, and estimating the slope inside each stratum gives -4.93 points per step on $[-6.51, -3.36]$ where the model can solve the problem and -4.36 on $[-6.54, -2.19]$ where it cannot. Both exclude zero, so chain depth carries information that difficulty does not account for.

3.3 Both results hold on a second domain and in other families

The findings above are measured on one benchmark and one model family, which two further sweeps address. A premise deletion probe built independently on MATH [Hendrycks et al., 2021], 818 parents and 1091 variants drawn from all seven subjects under the same two validity checks, reproduces the developmental ladder: decision quality runs 0.516, 0.557, 0.611 and 0.675 across the same checkpoints, against 0.509, 0.550, 0.690 and 0.763 on GSM8K. The shape is the same and the level is lower, which is expected, since competition problems are harder and a model declining one it simply cannot solve inflates the answerable rate and shrinks the measured gap. The depth decay reproduces there too, at -7.11 points per additional sentence on $[-9.87, -4.67]$ against a control slope containing zero.

Across families, three further instruct checkpoints graded the same GSM8K items. Decision quality reaches 0.795 for Qwen2.5 3B Instruct, 0.763 for Instella 3B Instruct, 0.759 for Qwen2.5 1.5B Instruct and 0.573 for OLMo-2 1B Instruct. Premise verification is therefore a property of instruction tuned small language models rather than of one pipeline, and Instella is competitive rather than exceptional. OLMo-2 is the informative case: its discrimination of +14.68 points on [+12.13, +17.23] clears zero comfortably, yet it declines 29.2 percent of answerable problems, so its decisions sit near chance. A difference of rates would call that a weaker version of the same capability; the decision metric identifies it as a different failure. Instella has the lowest wrong refusal rate of the four at 1.1 percent.

4 What exposure returns

4.1 Continued pretraining at controlled doses

Whether exposure supplies the answer is causal, and observation cannot settle it, so selected items were injected by continued pretraining at five doses under a fixed 8,388,608 token budget, held constant so a higher dose displaces filler rather than buying compute. Every arm carries positive controls evaluated before and after: injected document loss must fall and the model must continue an injected document verbatim, since a flat dose response with failed controls would measure a broken training run. Injection worked, verbatim reproduction climbing from 0.133 to 0.946 while held out loss rises, so the mixture taught those documents rather than the task.

4.2 The chain returns, the answer does not

Against a never injected control at the same dose, reproduction of the memorised calculation chain reaches +5.44 points on [+1.46, +9.38] at sixty four repetitions and rises with dose, while reproduction of the final answer reaches +1.52 on [-2.12, +5.20], an interval containing zero. The model reproduces the source text almost exactly and still does not recover the deleted quantity or the answer it determines. Repeating the identical mixture, budget, seed and probe on OLMo-2-1B, Qwen2.5-1.5B and Qwen2.5-3B gives +0.28, +0.00 and +2.28 points at the top dose, every interval containing zero, across fifteen dose points. Qwen2.5-3B is the sharpest case, driven to verbatim reproduction of 1.000 and still recovering the deleted answer no more often than a model that never saw the documents. Chain reproduction is established on Instella and underpowered elsewhere, as Appendix D sets out.

5 Discussion

Two capabilities that a single accuracy number conflates come apart here. Executing a solution procedure is present early and improves smoothly. Verifying that the premises support a solution at all appears later, improves sharply through instruction tuning, and degrades with the depth of the chain being tracked. Standard benchmark scoring cannot see the difference, because every item it scores is answerable. Reporting abstention under a prompt that permits it, against an answerable control, costs one extra evaluation pass and makes the second capability visible.

Limitations. One problem generator for the injection arm. Detection is measured through an explicit abstention option, so a model that recognises insufficiency but lacks the vocabulary to say so is scored as failing. Difficulty is stratified by the model’s own accuracy on intact problems rather than by an independent measure, and depth on MATH is sentence count rather than a verified calculation chain, which that benchmark does not annotate. Chain reproduction under injection is established on one model, and the replication is underpowered for it. Attribution to specific training documents is not established anywhere.

Conclusion. Measured against exposure fixed by containment rather than estimated, Instella’s grade school arithmetic performance is not recovered text: no test item approaches containment against the corpus, and heavy controlled injection returns the memorised calculation chain without returning the answer. What the probe reveals instead is a separable capability. The model detects that a problem no longer determines an answer, does so increasingly well across training, and loses that detection as the reasoning chain lengthens.

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A Related work

Huang et al. [Huang et al., 2025] construct hand authored hard perturbations of competition mathematics and attribute accuracy loss under an unchanged solution method to blind application of a learned procedure, a reading broadly consistent with the procedure level memorization in Section 3.2, established here against a corpus verified exactly rather than inferred. Wu et al. [Wu et al., 2025] show that reinforcement learning gains on mathematical reasoning depend on the pretraining exposure of the base policy and argue that clean synthetic problems are needed for reliable conclusions, a concern the checkpoint based design addresses by evaluating a policy before any relevant exposure occurred. Zhang et al. [Zhang et al., 2024] rebuild GSM8K from a contamination free procedure and report accuracy gaps of up to 13 points that track prior exposure across closed models, using an estimated exposure signal in the absence of any released training corpus; the present work substitutes an exact containment measurement for that estimate on a family where the corpus is public.

B Containment method, calibration, and corpus routes

The within split contamination contrast, high against low containment GSM8K train items, was closed rather than merely complicated by two findings. The corpus subset actually consumed during Instella 3B stage two pretraining, `train_119K`, is a 119,014 row subset of a 1,367,882 row released pool; containment computed against the pool over counts exposure, since 12.57 percent of GSM8K train reaches 0.80 or higher against the subset actually consumed against 23.34 percent against the pool, both measured on the same 891 parent problems through one extraction path. No GSM8K test item reaches containment 0.50 against either corpus, and the maximum over the full pool is 0.487, so the released corpus is train derived and the split remains a usable exposure contrast. Independently, `allenai/tulu-3-sft-mixture` is itself part of the Instella stage two pretraining mixture and `nvidia/OpenMathInstruct-2` enters at supervised fine tuning; both are generated in part from GSM8K train, and both carry it at containment 0.999 or higher on more than 97 percent of items. Every GSM8K train item therefore entered post stage one training through at least one of these routes regardless of its containment in the Instella specific synthetic corpus, so a high against low contrast defined on that corpus alone would compare two groups that are both almost entirely exposed once these mixtures are accounted for; the contrast is not attenuated by this, it is unidentifiable.

Item text is normalized to lowercase alphanumeric tokens and decomposed into every window of 13 consecutive tokens, following the overlap convention of Brown et al. [Brown et al., 2020]. Containment against a document is the fraction of an item’s windows found in that document; containment against a corpus is the maximum of that fraction over every document, computed in one streaming pass with an inverted index from window to source item so cost scales linearly in corpus size. Because the corpus is generated only from the GSM8K training split, every GSM8K test item is a verified negative, and the false positive rate is measured rather than assumed: at the adopted threshold of 0.80, zero of 1319 test items are flagged, a Wilson 95 percent upper bound of 0.29 percent; a looser rule flagging any window match flags 3 of 1000 sampled test items, 0.30 percent.

Within the Tulu 3 mixture, a sampled prefix of 300,000 rows, six of the mixture’s subsets, finds GSM8K train items reaching containment 0.999 through the `flan_v2_converted` subset, 6021 of 7473 items from 89,982 rows of that source, and through `wildchat` conversation logs, 4 items from 100,000 rows, an organic route in which users pasted benchmark problems into chat. Every scan reports rows examined alongside items found, since a sampled prefix bounds containment from below and a zero means absent from the rows scanned rather than absent from the corpus.

C The irrelevance marker

The inserted frame in the crossed design ends "and has no bearing on this question", which is an instruction to ignore the insertion carried inside the insertion itself. Shi et al. [Shi et al., 2023] report that such an instruction mitigates the loss irrelevant context causes, so the modest cost measured in Section 3.3 may sit inside their mitigated regime rather than in tension with it. Testing that requires removing the clause and nothing else. The replacement, "on the afternoon of the same day", holds the sentence at exactly twenty words and preserves position, topic noun and unused integer.

Table 2: Accuracy against each problem’s own original, and the effect of removing the irrelevance marker. Cluster bootstrap over parent problems, 4000 draws, 1764 parents.

Checkpoint	Marked	Unmarked	Marker effect
Instella 3B	−2.21 [−3.88, −0.51]	−5.67 [−7.43, −3.77]	− 3.46 [−4.90, −2.01]
SFT	+0.09 [−1.53, +1.67]	+0.11 [−1.53, +1.81]	+0.03 [−1.33, +1.36]
Instruct	−2.21 [−3.71, −0.65]	−2.78 [−4.34, −1.10]	−0.57 [−1.93, +0.79]

The marker governs the cost at the base stage two checkpoint and not after post training. One checkpoint of three is a single result rather than a pattern, and it is reported as such: it accounts for the size of the stage two effect and does not account for the residual two points at Instruct, which neither the number, nor topicality, nor the marker explains.

D The two withdrawn checks, and the dose response in full

The clause ladder is an externally authored perturbation suite whose rungs share template identifiers, which invites pairing each perturbed item against its own template head. The instances at each rung are independently generated rather than derived from one another, so a later rung is a different and harder problem set rather than the same problems with clauses appended, and an accuracy difference across rungs therefore mixes added content with added difficulty. Premise reordering permutes the non question sentences of a problem and preserves the gold answer by construction, but a narrative word problem often carries pronouns, temporal connectives and back references across sentences, so a permutation can leave text that no reader could solve. Filtering to items where no premise after the first carries such a dependency retains 24 percent of the set, and on that subset the effect is +2.2 points on [−6.5, +10.9] over 46 problems. Manual inspection finds violations that survive the filter, so even that residual is a spot check rather than an estimate.

E Reproduction

All generation, containment, and analysis code, together with every scored generation file and containment record underlying this work, is released for audit at a public repository and a paired dataset mirror, both anonymized for review and to be de anonymized on acceptance. Every containment number reported uses the maximum over individual documents definition stated above; an earlier internal computation of the same quantity against the corpus pooled as one text produced a materially different figure for the same released split and is superseded by the document level definition throughout this work. Random seeds are fixed across item selection, deletion variant construction, the continued pretraining mixture, and every cluster bootstrap. An assistant was used to help implement generation, containment, and analysis code under the direction of the authors, who independently verified every reported number against the underlying generation files; the assistant did not design the probes or select which results to report.

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